

rdi_validated_voters

scalars_rdi_validated_all_voting_ensemble

key	value
n_samples	3282
main_kwargs_applied.substitute	True
main_kwargs_applied.normalize_final_letters	True
timings_seconds.cross_wer	8.2
timings_seconds.ranking	126.8
timings_seconds.model_filtration	4752.1
timings_seconds.rank_distribution	5665.2
ranking_stats.n_models	9
ranking_stats.pool	seamless_m4t+whisper_v3+whisper_v2+fastconformer+xlsr+w2vbert+hubert+wav2vec_sb+cohere
rank_distribution_summary.n_subpools	382
rank_distribution_summary.n_boot_inner	1000
rank_distribution_summary.n_fusions	466
rank_distribution_summary.n_scorings	1971
rank_distribution_summary.seconds_fuse_and_score	880.3
rank_distribution_summary.seconds_bootstrap	32.7
rank_distribution_summary.weighting	filtration
rank_distribution_summary.exponent	1
rank_distribution_summary.mean_entropy	0.382
rank_distribution_summary.max_entropy	0.7219
rank_distribution_summary.n_models_modal_differs_from_full	1

ranking_rdi_validated_all_voting_ensemble

model	role	n_samples	wer_pseudo_loo	cer_pseudo_loo	rank_pseudo
whisper_v3	voter	3282	19.6085	9.4835	1
cohere	voter	3282	20.4038	12.2193	2
fastconformer	voter	3282	22.166	10.9467	3
whisper_v2	voter	3282	22.8449	11.1904	4
seamless_m4t	voter	3282	25.5763	13.9344	5
hubert	voter	3282	48.3265	19.7168	6
w2vbert	voter	3282	65.7823	26.146	7
xlsr	voter	3282	76.962	34.402	8
wav2vec_sb	voter	3282	81.0194	40.3413	9

model_confidence_rdi_validated_all_voting_ensemble

model	rank_pseudo_full	rank_modal	p_modal	p_at_full_rank	entropy_norm	rank_mean	rank_sd	rank_ci_lo	rank_ci_hi	agrees_with_full
whisper_v3	1	1	0.534	0.534	0.592	1.8994	1.2613	1	5	True
cohere	2	1	0.3471	0.28	0.7219	2.4057	1.4524	1	6	False
fastconformer	3	3	0.3652	0.3652	0.6505	3.0345	1.038	1	5	True
whisper_v2	4	4	0.384	0.384	0.7017	3.3446	1.1855	1	6	True
seamless_m4t	5	5	0.7037	0.7037	0.4571	4.5657	0.9706	2	6	True
hubert	6	6	0.8957	0.8957	0.1842	5.8275	0.5435	4	6	True
w2vbert	7	7	0.9653	0.9653	0.0791	6.9436	0.3109	6	7	True
xlsr	8	8	0.9874	0.9874	0.0326	7.9861	0.128	8	8	True
wav2vec_sb	9	9	0.9929	0.9929	0.0191	8.9929	0.0837	9	9	True

rank_distribution_rdi_validated_all_voting_ensemble

model	rank_1	rank_2	rank_3	rank_4	rank_5	rank_6	rank_7	rank_8	rank_9
seamless_m4t	6.3447e-05	0.0415	0.1449	0.0741	0.7037	0.0219	0.0113	0.001	0.0017
whisper_v3	0.534	0.2383	0.0953	0.0849	0.0294	0.0138	0.0022	0.0008	0.0012
whisper_v2	0.0682	0.1769	0.2525	0.384	0.091	0.0212	0.0035	0.0015	0.0012
fastconformer	0.0506	0.2633	0.3652	0.2623	0.0443	0.0116	0.0013	0.0002	0.0012
xlsr	0	0	0	0	0	0.0013	0.0112	0.9874	0
w2vbert	0	0	0	0	0.0217	0.013	0.9653	0	0
hubert	0	0	0	0.0718	0.0315	0.8957	0	0.0006	0.0004
wav2vec_sb	0	0	0	0	0	0	0	0.0071	0.9929
cohere	0.3471	0.28	0.1421	0.1229	0.0785	0.0215	0.0052	0.0015	0.0012

pair_probability_rdi_validated_all_voting_ensemble

pair	pred_winner_full	gap_wer_pseudo	p_first_beats_second	p_supports_full
whisper_v2 fastconformer	fastconformer	0.6789	0.3844	0.6156
whisper_v3 cohere	whisper_v3	0.7953	0.6177	0.6177
fastconformer cohere	cohere	1.7622	0.3457	0.6543
whisper_v2 cohere	cohere	2.4411	0.2535	0.7465
whisper_v3 whisper_v2	whisper_v3	3.2364	0.803	0.803
whisper_v3 fastconformer	whisper_v3	2.5575	0.8169	0.8169
seamless_m4t cohere	cohere	5.1725	0.1384	0.8616
seamless_m4t fastconformer	fastconformer	3.4103	0.1308	0.8692
seamless_m4t whisper_v2	whisper_v2	2.7314	0.1304	0.8696
seamless_m4t whisper_v3	whisper_v3	5.9678	0.0895	0.9105
whisper_v3 hubert	whisper_v3	28.718	0.9634	0.9634
fastconformer hubert	fastconformer	26.1605	0.9635	0.9635

pair	pred_winner_full	gap_wer_pseudo	p_first_beats_second	p_supports_full
hubert cohere	cohere	27.9227	0.0365	0.9635
whisper_v2 hubert	whisper_v2	25.4816	0.9636	0.9636
seamless_m4t hubert	seamless_m4t	22.7502	0.9636	0.9636
seamless_m4t w2vbert	seamless_m4t	40.206	0.986	0.986
w2vbert cohere	cohere	45.3785	0.0099	0.9901
fastconformer w2vbert	fastconformer	43.6163	0.991	0.991
whisper_v2 w2vbert	whisper_v2	42.9374	0.9911	0.9911
w2vbert hubert	hubert	17.4558	0.0084	0.9916
whisper_v3 w2vbert	whisper_v3	46.1738	0.9924	0.9924
xlsr cohere	cohere	56.5582	0.0027	0.9973
whisper_v2 xlsr	whisper_v2	54.1171	0.9973	0.9973
seamless_m4t xlsr	seamless_m4t	51.3857	0.9973	0.9973
whisper_v3 xlsr	whisper_v3	57.3535	0.9979	0.9979
seamless_m4t wav2vec_sb	seamless_m4t	55.4431	0.9983	0.9983
fastconformer xlsr	fastconformer	54.796	0.9986	0.9986
xlsr w2vbert	w2vbert	11.1797	0.0013	0.9987
whisper_v3 wav2vec_sb	whisper_v3	61.4109	0.9988	0.9988
fastconformer wav2vec_sb	fastconformer	58.8534	0.9988	0.9988
wav2vec_sb cohere	cohere	60.6156	0.0012	0.9988
whisper_v2 wav2vec_sb	whisper_v2	58.1745	0.9988	0.9988
xlsr hubert	hubert	28.6354	0.0011	0.9989
hubert wav2vec_sb	hubert	32.6929	0.9996	0.9996
xlsr wav2vec_sb	xlsr	4.0574	1	1
w2vbert wav2vec_sb	w2vbert	15.2371	1	1

model_filtration_rdi_validated_all_voting_ensemble

model	n_subpools	f_bar	f_sd	f_min	f_max	survival
whisper_v3	219	0.0287	0.0404	0.0052	0.3434	0.9713
fastconformer	219	0.0345	0.0517	0.0055	0.3501	0.9655
cohere	219	0.0353	0.0345	0.0128	0.2377	0.9647
whisper_v2	219	0.0461	0.0561	0.0146	0.4281	0.9539
seamless_m4t	219	0.0471	0.046	0.0152	0.3193	0.9529
hubert	219	0.258	0.1801	0.0615	0.8668	0.742
w2vbert	219	0.5841	0.158	0.266	0.9458	0.4159
xlsr	219	0.8171	0.1213	0.3894	0.9872	0.1829
wav2vec_sb	219	0.877	0.0916	0.5271	0.9909	0.123

by_pool_size_rdi_validated_all_voting_ensemble

k	pools	total_weight	mean_entropy	max_entropy
4	126	0.2557	0.0381	0.2502
5	126	0.3196	0.0413	0.2512
6	84	0.2557	0.0473	0.1937
7	36	0.1279	0.0352	0.18
8	9	0.0365	0.0197	0.0428
9	1	0.0046	0.004	0.004

cross_wer_rdi_validated_all_voting_ensemble

	seamless_m4t	whisper_v3	whisper_v2	fastconformer	xlsr	w2vbert	hubert	wav2vec_sb	cohere
seamless_m4t	0	29.36	31.81	29.96	80.67	69.84	54.49	83.84	26.7
whisper_v3	29.36	0	21.8	26.02	77.79	67.26	50.08	81.92	24.08
whisper_v2	31.81	21.8	0	28.69	77.9	67.68	50.58	81.9	27.01
fastconformer	29.96	26.02	28.69	0	77.97	67.4	50.2	81.85	24.25
xlsr	80.67	77.79	77.9	77.97	0	84.08	84.11	80.99	90.49
w2vbert	69.84	67.26	67.68	67.4	84.08	0	75.39	86.67	77.96
hubert	54.49	50.08	50.58	50.2	84.11	75.39	0	80.85	54.17
wav2vec_sb	83.84	81.92	81.9	81.85	80.99	86.67	80.85	0	98.9
cohere	26.7	24.08	27.01	24.25	90.49	77.96	54.17	98.9	0

subpool_filtration_rdi_validated_all_voting_ensemble (first 60 of 1971 rows)

pool	k	model	n_samples	filtered_out	filtered_out_rate
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	seamless_m4t	3282	1048	0.3193
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	whisper_v3	3282	635	0.1935
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	whisper_v2	3282	994	0.3029
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	fastconformer	3282	1136	0.3461
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	seamless_m4t	3282	128	0.039
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	whisper_v3	3282	23	0.007
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	whisper_v2	3282	58	0.0177
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	xlsr	3282	3133	0.9546
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	seamless_m4t	3282	229	0.0698
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	whisper_v3	3282	87	0.0265
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	whisper_v2	3282	159	0.0484
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	w2vbert	3282	2936	0.8946
seamless_m4t+whisper_v3+whisper_v2+hubert	4	seamless_m4t	3282	461	0.1405
seamless_m4t+whisper_v3+whisper_v2+hubert	4	whisper_v3	3282	188	0.0573

pool	k	model	n_samples	filtered_out	filtered_out_rate
seamless_m4t+whisper_v3+whisper_v2+hubert	4	whisper_v2	3282	330	0.1005
seamless_m4t+whisper_v3+whisper_v2+hubert	4	hubert	3282	2600	0.7922
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	seamless_m4t	3282	93	0.0283
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	whisper_v3	3282	22	0.0067
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	whisper_v2	3282	48	0.0146
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	wav2vec_sb	3282	3159	0.9625
seamless_m4t+whisper_v3+whisper_v2+cohere	4	seamless_m4t	3282	1021	0.3111
seamless_m4t+whisper_v3+whisper_v2+cohere	4	whisper_v3	3282	759	0.2313
seamless_m4t+whisper_v3+whisper_v2+cohere	4	whisper_v2	3282	1121	0.3416
seamless_m4t+whisper_v3+whisper_v2+cohere	4	cohere	3282	780	0.2377
seamless_m4t+whisper_v3+fastconformer+xlsr	4	seamless_m4t	3282	113	0.0344
seamless_m4t+whisper_v3+fastconformer+xlsr	4	whisper_v3	3282	56	0.0171
seamless_m4t+whisper_v3+fastconformer+xlsr	4	fastconformer	3282	52	0.0158
seamless_m4t+whisper_v3+fastconformer+xlsr	4	xlsr	3282	3157	0.9619
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	seamless_m4t	3282	200	0.0609
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	whisper_v3	3282	147	0.0448
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	fastconformer	3282	138	0.042
seamless_m4t+whisper_v3+fastconformer+w2vbert	4	w2vbert	3282	2954	0.9001
seamless_m4t+whisper_v3+fastconformer+hubert	4	seamless_m4t	3282	380	0.1158
seamless_m4t+whisper_v3+fastconformer+hubert	4	whisper_v3	3282	330	0.1005
seamless_m4t+whisper_v3+fastconformer+hubert	4	fastconformer	3282	300	0.0914
seamless_m4t+whisper_v3+fastconformer+hubert	4	hubert	3282	2617	0.7974
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	seamless_m4t	3282	82	0.025
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	whisper_v3	3282	43	0.0131
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	fastconformer	3282	36	0.011
seamless_m4t+whisper_v3+fastconformer+wav2vec_sb	4	wav2vec_sb	3282	3162	0.9634
seamless_m4t+whisper_v3+fastconformer+cohere	4	seamless_m4t	3282	954	0.2907
seamless_m4t+whisper_v3+fastconformer+cohere	4	whisper_v3	3282	1127	0.3434
seamless_m4t+whisper_v3+fastconformer+cohere	4	fastconformer	3282	1072	0.3266
seamless_m4t+whisper_v3+fastconformer+cohere	4	cohere	3282	607	0.1849
seamless_m4t+whisper_v3+xlsr+w2vbert	4	seamless_m4t	3282	125	0.0381
seamless_m4t+whisper_v3+xlsr+w2vbert	4	whisper_v3	3282	52	0.0158
seamless_m4t+whisper_v3+xlsr+w2vbert	4	xlsr	3282	2603	0.7931
seamless_m4t+whisper_v3+xlsr+w2vbert	4	w2vbert	3282	1676	0.5107
seamless_m4t+whisper_v3+xlsr+hubert	4	seamless_m4t	3282	116	0.0353
seamless_m4t+whisper_v3+xlsr+hubert	4	whisper_v3	3282	67	0.0204
seamless_m4t+whisper_v3+xlsr+hubert	4	xlsr	3282	2779	0.8467
seamless_m4t+whisper_v3+xlsr+hubert	4	hubert	3282	542	0.1651
seamless_m4t+whisper_v3+xlsr+wav2vec_sb	4	seamless_m4t	3282	124	0.0378
seamless_m4t+whisper_v3+xlsr+wav2vec_sb	4	whisper_v3	3282	69	0.021

pool	k	model	n_samples	filtered_out	filtered_out_rate
seamless_m4t+whisper_v3+xlsr+wav2vec_sb	4	xlsr	3282	2419	0.7371
seamless_m4t+whisper_v3+xlsr+wav2vec_sb	4	wav2vec_sb	3282	2638	0.8038
seamless_m4t+whisper_v3+xlsr+cohere	4	seamless_m4t	3282	92	0.028
seamless_m4t+whisper_v3+xlsr+cohere	4	whisper_v3	3282	53	0.0161
seamless_m4t+whisper_v3+xlsr+cohere	4	xlsr	3282	3155	0.9613
seamless_m4t+whisper_v3+xlsr+cohere	4	cohere	3282	49	0.0149

subpool_ranks_rdi_validated_all_voting_ensemble (first 60 of 4731 rows)

pool	k	model	rank	count
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	seamless_m4t	5	1000
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	whisper_v3	1	50
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	whisper_v3	2	950
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	whisper_v2	3	1000
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	fastconformer	4	1000
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	xlsr	8	1000
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	w2vbert	7	1000
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	hubert	6	1000
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	wav2vec_sb	9	1000
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	cohere	1	950
seamless_m4t+whisper_v3+whisper_v2+fastconformer	4	cohere	2	50
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	seamless_m4t	5	1000
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	whisper_v3	2	3
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	whisper_v3	3	997
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	whisper_v2	4	1000
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	fastconformer	2	997
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	fastconformer	3	3
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	xlsr	8	1000
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	w2vbert	7	1000
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	hubert	6	1000
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	wav2vec_sb	9	1000
seamless_m4t+whisper_v3+whisper_v2+xlsr	4	cohere	1	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	seamless_m4t	5	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	whisper_v3	2	7
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	whisper_v3	3	993
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	whisper_v2	4	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	fastconformer	2	993
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	fastconformer	3	7
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	xlsr	8	1000

pool	k	model	rank	count
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	w2vbert	7	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	hubert	6	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	wav2vec_sb	9	1000
seamless_m4t+whisper_v3+whisper_v2+w2vbert	4	cohere	1	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	seamless_m4t	5	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	whisper_v3	2	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	whisper_v2	3	59
seamless_m4t+whisper_v3+whisper_v2+hubert	4	whisper_v2	4	941
seamless_m4t+whisper_v3+whisper_v2+hubert	4	fastconformer	3	941
seamless_m4t+whisper_v3+whisper_v2+hubert	4	fastconformer	4	59
seamless_m4t+whisper_v3+whisper_v2+hubert	4	xlsr	8	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	w2vbert	7	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	hubert	6	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	wav2vec_sb	9	1000
seamless_m4t+whisper_v3+whisper_v2+hubert	4	cohere	1	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	seamless_m4t	5	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	whisper_v3	3	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	whisper_v2	4	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	fastconformer	2	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	xlsr	8	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	w2vbert	7	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	hubert	6	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	wav2vec_sb	9	1000
seamless_m4t+whisper_v3+whisper_v2+wav2vec_sb	4	cohere	1	1000
seamless_m4t+whisper_v3+whisper_v2+cohere	4	seamless_m4t	5	1000
seamless_m4t+whisper_v3+whisper_v2+cohere	4	whisper_v3	1	1000
seamless_m4t+whisper_v3+whisper_v2+cohere	4	whisper_v2	3	72
seamless_m4t+whisper_v3+whisper_v2+cohere	4	whisper_v2	4	928
seamless_m4t+whisper_v3+whisper_v2+cohere	4	fastconformer	2	19
seamless_m4t+whisper_v3+whisper_v2+cohere	4	fastconformer	3	909
seamless_m4t+whisper_v3+whisper_v2+cohere	4	fastconformer	4	72